Finding Slope — ANSWER KEY

Examples 1 - 4:

Find the slope, simplify (reduce) if needed.

Given the points (1, 1) and (5, 7)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{7 - 1}{5 - 1} = 6/4 = 3/2$$

Find the slope, simplify (reduce) if needed.

Given the points (-4, -3) and (2, -6)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-6 - (-3)}{2 - (-4)} = -3/6 = -1/2$$

Find the slope, simplify (reduce) if needed.

Given the points (-4, 3) and (5, 3)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{3 - 3}{5 - (-4)} = 0$$

Find the slope, simplify (reduce) if needed.

Given the points (-2, -5) and (-2, 4)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4 - (-5)}{-2 - (-2)} = \text{undefined}$$

Find the slope from each pair of points. Simplify if needed.

1. Find the slope, simplify (reduce) if needed.

Given the points (1, 3) and (5, 11)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{11 - 3}{5 - 1} = 8/4 = 2$$

2. Find the slope, simplify (reduce) if needed.

Given the points (-2, 1) and (3, -4)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-4 - 1}{3 - (-2)} = -5/5 = -1$$

3. Find the slope, simplify (reduce) if needed.

Given the points (0, -3) and (4, 3)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{3 - (-3)}{4 - 0} = 6/4 = 3/2$$

4. Find the slope, simplify (reduce) if needed.

Given the points (-4, 2) and (2, -1)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-1 - 2}{2 - (-4)} = -3/6 = -1/2$$

5. Find the slope, simplify (reduce) if needed.

Given the points (2, 5) and (7, 9)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{9 - 5}{7 - 2} = 4/5$$

6. Find the slope, simplify (reduce) if needed.

Given the points (-3, -2) and (3, 2)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{2 - (-2)}{3 - (-3)} = 4/6 = 2/3$$

7. Find the slope, simplify (reduce) if needed.

Given the points (1, 4) and (6, 4)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4 - 4}{6 - 1} = 0$$

8. Find the slope, simplify (reduce) if needed.

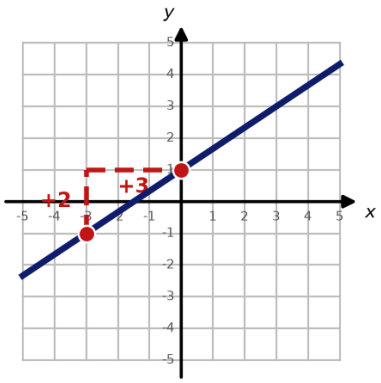
Given the points (-2, 5) and (-2, -3)

$$\frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{-3 - 5}{-2 - (-2)} = \text{undefined}$$

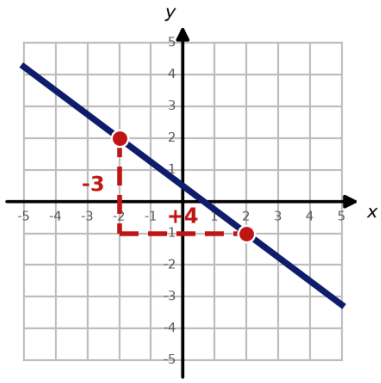
Find the slope of each line from the graph.

9. Find the slope of the line.



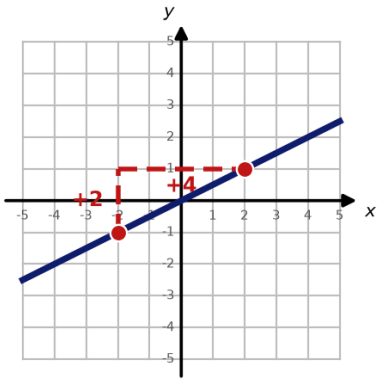
$m = \frac{2}{3} = 2/3$

10. Find the slope of the line.



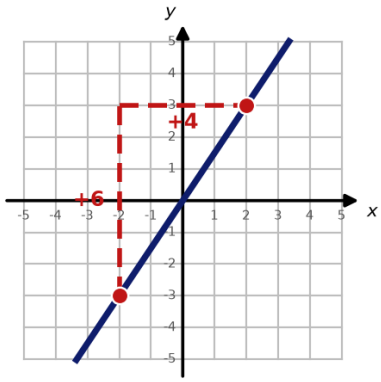
$m = \frac{-3}{4} = -3/4$

11. Find the slope of the line.



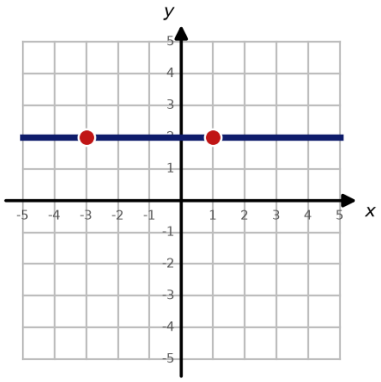
$m = \frac{2}{4} = 1/2$

12. Find the slope of the line.



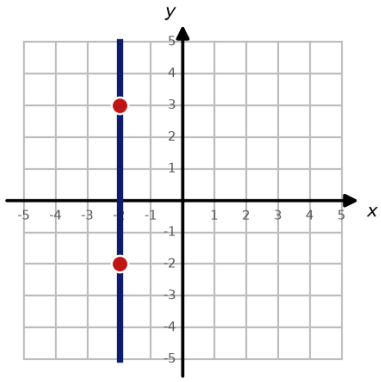
$m = \frac{6}{4} = 3/2$

13. Find the slope of the line.



$m = \frac{0}{4} = 0$

14. Find the slope of the line.



$m = \frac{-5}{0} =$ undefined